The Cost of Reproducing Learning Trajectories: Sharp Resource Bounds for Adaptive Resistor Networks

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Abstract

A resistor network's instantaneous electrical equivalent can fail to reproduce its response during learning. We determine the minimum number of trainable edges needed to reproduce every admissible learning trajectory of N parallel three-ray stars. Source and replacement follow the same conductance-weighted gradient law, with their own output errors, identical targets and physical clock, a common initial conductance budget, and no auxiliary memory. For sufficiently small positive tolerance δ, uniformly over source preparations and target histories on a fixed finite horizon, the minimum has order

$$\min\{N, [1 + \log(1/\delta)]^2\}.$$

The lower bound allows arbitrary passive replacement topologies and discontinuous preparation; a fixed star ensemble with one smooth positive initializer attains the upper bound. Exact reproduction requires $3N$ edges in the worst case, also at sufficiently small positive tolerance for each fixed N. These asymptotic bounds concern reproduction of learning trajectories, rather than reaching the same trained response by any path. They follow from uniform electrical sensitivity, finite training experiments that distinguish source states, and smooth physical realization of analytic response features.

Keywords: physical learning; adaptive resistor networks; nonlinear dynamics; model reduction; resource bounds; observability; metric entropy.

1 Introduction

In a physical learning network, the elements that transmit signals also store the changes produced by training. Coupled-learning theory shows how local physical responses can guide those changes [1], and experiments with variable resistors demonstrate learning through distributed electrical updates [2]. Reducing such a network raises a dynamical question: how much of its internal structure must a smaller device retain to reproduce its response to future training?

Static circuit equivalence gives a particularly simple starting point. Three effective conductances determine the entire instantaneous boundary response of a passive three-terminal network, so a triangle can reproduce it exactly [3]. That replacement can nevertheless change the learning dynamics. In the example of Figure 1 and Section 2.3, a symmetric star ensemble and its equivalent triangle have the same terminal response but differ by a factor of three in their initial response velocities under a common target. The question is how small a replacement can be while preserving the response throughout every allowed training history.

We determine the asymptotic answer for a normalized family of N parallel three-ray stars, containing $3N$ trainable conductances. Source and replacement share a conductance-weighted gradient law, target class, physical clock, and initial conductance budget; each uses its own output error. For sufficiently small uniform error δ on a fixed finite horizon, the necessary and sufficient number of trainable edges has order

$$\min\{N, [1 + \log(1/\delta)]^2\}. \tag{1.1}$$

The lower bound covers every admissible passive graph, even when its topology and initialization are selected discontinuously for each source state. One fixed star ensemble with a single smooth positive initializer attains the upper bound. Thus no other admissible topology has an advantage that grows with source size or required accuracy. At zero error, the worst-case minimum is exactly $3N$. These are asymptotic statements about edge count; the constants do not provide a practical device size or an experimentally useful accuracy threshold.

The squared logarithm has a geometric origin in the construction. A star has two shape coordinates and a conductance mass. Homogeneity makes its initial mass a weight, while the evolved response is uniformly analytic in its two initial shape coordinates. Degree-k approximation therefore gives exponentially small error using $D_k = J\binom{k+2}{2} = O(k^2)$ features, where J is a fixed number of local coordinate patches. Taking $k = O(1 + \log(1/\delta))$ explains the upper scale in Equation (1.1). Necessity requires a separate physical argument: finite admissible training histories distinguish enough variations in both shape coordinates to force the same order for every replacement graph. Analytic mixtures can otherwise admit more economical representations than a dimension count suggests [4].

Preserving training trajectories has direct precedents. Neural tangent transfer constructs sparse trainable models and proves output-trajectory equivalence in a linear setting [5]. Intrinsic training dynamics study closure on reduced variables and its dependence on initialization [6]. Whole-map approximation is also central to offline/online model-reduction benchmarks [7]. The contribution here is a matching necessary and sufficient physical resource bound over all admissible replacement topologies. The electrical update is an idealized model with local voltage factors and shared residual signals; its relation to coupled-learning devices is specified in Section 2.

The proof follows three steps. First, a graph-independent sensitivity bound limits how many future responses a small network can produce, making metric-entropy methods applicable to arbitrary initialization rules [8]. Second, a two-dimensional array of source shapes produces a sufficiently rich family of responses to finite training programs. Third, analytic approximation and a smooth positive packing construction realize the needed features in physical conductances,

using positive moment representations as a starting point [9, 10]. The exact result additionally uses mass variations acquired during training: varying preparation and an initial target sequence jointly produces an open $3N$-dimensional family distinguishable by subsequent boundary observations.

2 Physical model and admissible reductions

2.1 Electrical response and learning law

Let $G = (V, E)$ be a finite undirected resistor multigraph with three named boundary vertices $\partial V = \{1, 2, 3\}$. Every edge e has positive conductance g_e. We remove electrically irrelevant components and require each connected component of interior vertices to attach to a terminal, so that the interior Dirichlet block is invertible. Parallel edges are allowed and counted separately. All replacement edges are trainable; the class contains no auxiliary fixed-conductance edges, time-dependent circuit elements, or hidden state variables. Passivity refers to the electrical network at fixed conductances. The probes, error signals, and parameter adaptation are additional operations whose energy and circuitry are outside the edge count.

Let B be the incidence matrix for a chosen orientation. The graph has nodal Laplacian L_G and Dirichlet-to-Neumann (DtN) matrix Λ_G, given by

$$L_G(g) = B^\top \operatorname{diag}(g) B, \qquad \Lambda_G(g) = L_{\partial\partial} - L_{\partial I} L_{II}^{-1} L_{I\partial}.$$

If there are no interior vertices, $\Lambda_G = L_{\partial\partial}$. Symmetry and $\Lambda_G \mathbf{1} = 0$ imply that the three effective conductances

$$C_G(g) = \big(-\Lambda_{12}(g), -\Lambda_{13}(g), -\Lambda_{23}(g)\big) \in \mathbb{R}^3_+ \tag{2.1}$$

determine the entire static boundary response. The map C_G is smooth on positive conductances, positively homogeneous of degree one, and componentwise nonnegative.

Conductances and target responses are expressed in a fixed reference conductance unit. A common constant learning coefficient sets the time unit; neither unit changes with the network, state, or target. We study an idealized continuous adaptation law, with electrical equilibration treated as instantaneous on that time scale. For a target $q(t) \in \mathbb{R}^3$, the loss and learning law are

$$\mathcal{L}_q(g) = \tfrac{1}{2}\|C_G(g) - q\|_2^2, \qquad \dot{g}_e = -g_e\, \partial_{g_e} \mathcal{L}_q(g) = g_e \sum_{a=1}^{3} \partial_{g_e} C_{G,a}(g)\, [q_a - C_{G,a}(g)]. \tag{2.2}$$

The resulting terminal dynamics make the distinction from static equivalence explicit:

$$\dot{C}_G = K_G(g)(q - C_G), \qquad K_G(g) = DC_G(g)\, \operatorname{diag}(g)\, DC_G(g)^\top. \tag{2.3}$$

The current response C_G and the matrix K_G controlling its velocity depend differently on the conductances. Equality of C_G therefore need not preserve the next training step. Reproducing a whole trajectory requires this response equation to remain matched as the internal states evolve.

For each terminal a, impose a unit voltage there and zero at the other terminals, and let

v_e^a be the resulting edge voltage drop. Electrical sensitivity gives $\partial_{g_e} C_{ab} = -v_e^a v_e^b$ for $a < b$ (Equation (4.2)). Thus $\dot{g}_e/g_e = -\sum_{a<b}(q_{ab} - C_{ab}) v_e^a v_e^b$: the relative adaptation rate is a residual-weighted sum of local voltage products. In the absolute update, $g_e v_e^a v_e^b$ is a signed cross-power term. It is related to ordinary edge dissipation by

$$g_e v_e^a v_e^b = \tfrac{1}{2}\big(P_e^{a+b} - P_e^a - P_e^b\big), \qquad P_e^a = g_e (v_e^a)^2, \quad P_e^{a+b} = g_e (v_e^a + v_e^b)^2.$$

Here powers use the corresponding fixed voltage and conductance units. This identity connects adaptation to electrical measurements, while retaining the sign and the contribution of two different probes.

The model assumes access to three common boundary probes and a shared three-component output residual. Conductances are held fixed during the idealized probe measurements, and no source-specific state other than those conductances is retained between updates. Coupled-learning rules instead obtain local updates from a comparison of free and clamped states [1]; the twin-network experiment in Ref. [2] realizes that comparison in hardware. These works motivate physical learning with local electrical factors, but their update rules and measurement interfaces differ from Equation (2.2).

The conductance factor also makes the law consistent under parallel subdivision. If one edge is replaced by parallel edges with the same endpoints, their voltage drops and loss derivatives coincide. They have equal logarithmic rates, so their conductance fractions stay fixed and their sum follows the original edge. Geometrically, the mobility defines the weighted metric $\sum_e \mathrm{d}g_e^2/g_e$. The square-root coordinates $x_e = 2\sqrt{g_e}$ give

$$\sum_e \frac{\mathrm{d}g_e^2}{g_e} = \sum_e \mathrm{d}x_e^2, \qquad \dot{x} = -\nabla_x \mathcal{L}_q.$$

Electrical equilibrium then gives a sensitivity bound uniform over all positive conductances (Lemma 4.1).

Let $\mathcal{Q}$ contain all piecewise constant programs $q : [0,1] \to [0,1]^3$ with finitely many switches. Write $C^q_{G,g^0}(t)$ for the response from initial state g^0. The object to be reproduced is the full family

$$\mathcal{R}_G(g^0) : \ (q,t) \longmapsto C^q_{G,g^0}(t), \qquad \|\mathcal{R}_G(g^0) - \mathcal{R}_H(z^0)\|_\infty = \sup_{q\in\mathcal{Q}} \sup_{0\le t\le 1} \|C^q_{G,g^0}(t) - C^q_{H,z^0}(t)\|_\infty. \tag{2.4}$$

Each program defines a separate experiment, with the source restarted at g^0 and the replacement at z^0. The norm compares the two responses throughout every such experiment.

2.2 The source family

For three positive conductances $y = (y_1, y_2, y_3)$, a star has effective conductances

$$S(y) = \frac{1}{y_1 + y_2 + y_3}\,(y_1y_2, y_1y_3, y_2y_3). \tag{2.5}$$

For $N \geq 2$, let G_N consist of N stars in parallel, with common boundary vertices and distinct internal centers. The $3N$ conductances are independently trainable. The initial shape tuple $t = (t_1, \ldots, t_N)$ ranges over

$$\Delta = \Big\{x \in \mathbb{R}^3 : \; x_j \geq \tfrac{1}{6}, \; \sum_{j=1}^{3} x_j = 1\Big\}, \qquad g_i^0 = \frac{t_i}{N}, \quad t_i \in \Delta. \tag{2.6}$$

Every source starts with total conductance $\sum_{i,e} g_{ie}^0 = 1$, and its response is $C_{G_N}(g) = \sum_i S(g_i)$. Preparation varies over the entire compact family Δ^N. Each star begins with conductance mass $1/N$; both its mass and its normalized shape subsequently evolve under Equation (2.2).

2.3 A static equivalent with a different learning rate

Set every source shape to $(1/3, 1/3, 1/3)$, so every source edge initially has conductance $1/(3N)$. Its N stars give $C_0 = (1/9, 1/9, 1/9)$. The triangle with all three conductances $1/9$ has exactly the same response and uses total conductance $1/3$, within the allowed unit budget. At any symmetric star,

$$DS = \frac{1}{9}\begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix}, \qquad DS\,DS^{\top} = \frac{1}{9} I_3.$$

Summing the N contributions to Equation (2.3) gives

$$K_{G_N}(g^0) = N\frac{1}{3N} DS\,DS^{\top} = \frac{1}{27} I_3, \qquad K_{\triangle}(z^0) = \frac{1}{9} I_3. \tag{2.7}$$

For any target $q \neq C_0$, the triangle's initial response velocity is three times that of the star ensemble, as summarized in Figure 1. This example isolates the failure of static reduction. The identical source stars can themselves be merged into one star of mass one, which preserves all subsequent responses by homogeneity. The lower bound growing with N instead concerns the full family of differing initial shapes.

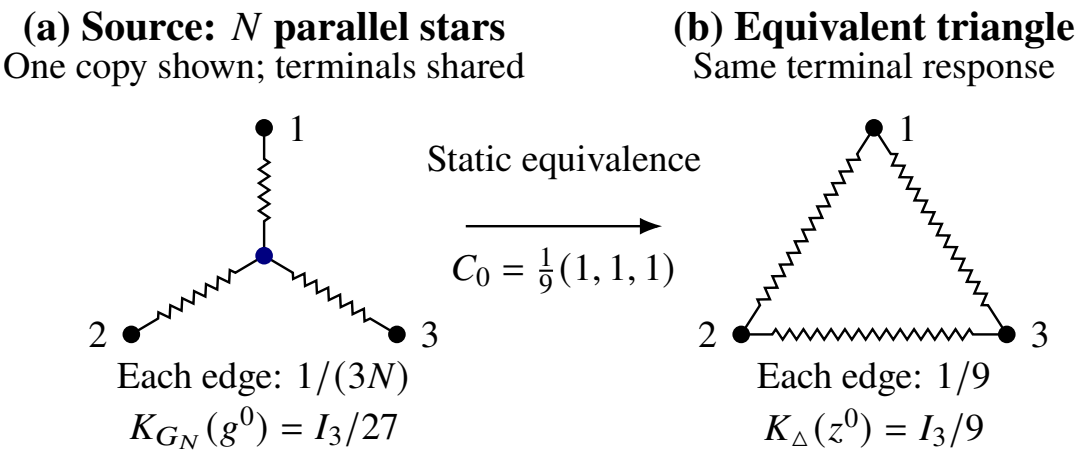


Figure 1: Static equivalence and learning response. All source stars are initially symmetric, with conductance $1/(3N)$ on each edge. Connecting their corresponding terminals gives the same complete response C_0 as the triangle. Under the common learning law, $\dot{C} = K(q - C)$, the triangle has three times the source's initial response velocity for every $q \neq C_0$. This is an exact calculation, not a simulated trajectory. Identical stars can be merged into one star, so the example illustrates failure of the static triangle, not the worst-case resource lower bound.

2.4 Replacement classes

A replacement graph H has $m = |E(H)|$ edges, the same three terminals, positive initial conductances z^0, and initial budget $\sum_e z_e^0 \leq 1$. It follows Equation (2.2) with its own error $q - C_H(z)$. Initialization takes place before the future target program is known. After preparation, the graph stays fixed and the only source-dependent continuous state retained by the device is its conductance vector.

We compare three classes at the same error tolerance:

(i) *Free preparation*, $m_{\text{free}}(N, \delta)$: both the graph and its positive initial conductances may depend arbitrarily, including discontinuously, on $t \in \Delta^N$.

(ii) *Smooth preparation*, $m_{\text{all}}(N, \delta)$: one graph serves all of Δ^N, with a single C^1 initialization map $E : \Delta^N \to (0, \infty)^m$, extendible to a relative neighborhood, and satisfying $\sum_e E_e(t) \leq 1$.

(iii) *Smooth star preparation*, $m_{\text{star}}(N, \delta)$: smooth preparation is restricted to a fixed parallel ensemble of three-ray stars.

For each class, the minimum is the least edge count satisfying

$$\sup_{t \in \Delta^N} \|\mathcal{R}_{G_N}(g^0(t)) - \mathcal{R}_H(z^0(t))\|_\infty \leq \delta. \tag{2.8}$$

Only free preparation allows H to vary with the source initialization. The error includes the initial response, since the time interval in Equation (2.4) contains zero. Preparation has no computational or precision budget; smooth preparation adds only the stated regularity requirement. The optimized resource counts trainable edges. Sensing, communication, computation, material

volume, and manufacturing tolerance are separate costs. The nested classes and the unchanged source give

$$m_{\mathrm{free}} \le m_{\mathrm{all}} \le m_{\mathrm{star}} \le 3N. \tag{2.9}$$

3 Resource bounds

Theorem 3.1 (Sharp finite-accuracy resource law)**.** *There exist constants $c, C, \delta_0 > 0$, depending only on the fixed model in Section 2, such that for every $N \ge 2$ and $0 < \delta \le \delta_0$, with $L_\delta = 1 + \log(1/\delta)$,*

$$c \min\{N, L_\delta^2\} \le m_{\mathrm{free}}(N, \delta) \le m_{\mathrm{all}}(N, \delta) \le m_{\mathrm{star}}(N, \delta) \le C \min\{N, L_\delta^2\}. \tag{3.1}$$

The upper bound is attained by a fixed parallel star ensemble with a single C^∞ initialization map and strictly positive conductances. The lower bound allows discontinuous, source-dependent choices of topology and initialization.

Corollary 3.2 (Constant-factor comparison of topologies)**.** *For the same N and δ as in Theorem 3.1,*

$$m_{\mathrm{star}}(N, \delta) \le (C/c)\, m_{\mathrm{free}}(N, \delta). \tag{3.2}$$

The comparison uses the same N and tolerance on both sides, although the free-preparation class allows a separate topology and arbitrary initialization for each source state.

Theorem 3.3 (Exact trajectory saturation)**.** *For every $N \ge 2$,*

$$m_{\mathrm{free}}(N, 0) = m_{\mathrm{all}}(N, 0) = m_{\mathrm{star}}(N, 0) = 3N. \tag{3.3}$$

For each fixed N there is also a $\delta_N > 0$ such that all three minima equal $3N$ whenever $0 < \delta \le \delta_N$.

Theorem 3.1 is proved in Sections 4 to 6, and Theorem 3.3 in Section 7. Together they describe two limits. At fixed positive tolerance, the edge count remains bounded as the source size grows. At fixed source size N, letting $\delta \downarrow 0$ eventually requires exactly $3N$ edges by Theorem 3.3. This exact coefficient follows from the saturation theorem; the order estimate in Theorem 3.1 alone gives only $\Theta(N)$ in that regime. The proof gives neither a numerical value of δ_N nor a lower bound on it uniform in N.

Table 1 contrasts this dynamic resource with the three conductances sufficient for the instantaneous response.

4 Graph-independent sensitivity and covering estimates

A replacement may be prepared by an arbitrary rule and may change its topology with the source initialization. To obtain a lower bound for this entire class, we estimate how many distinct response maps its conductances can produce. The first step is a sensitivity bound that is uniform over graph topology and remains valid when some conductances are very small.

Table 1: Three levels of reproduction for the declared source family. Dynamic edge counts are worst-case minima over the entire preparation family, with all admissible target histories and the same unit horizon. Here $L_\delta = 1 + \log(1/\delta)$. The static triangle is a separate instantaneous comparator. For fixed $0 < \delta \le \delta_0$, the dynamic edge count is bounded as $N \to \infty$. For fixed N, Theorem 3.3 gives exact saturation at $3N$ as $\delta \downarrow 0$; the Θ law alone gives only the order N. The positive threshold δ_N is not evaluated; it may depend on N.

Response to preserve	Accuracy	Trainable edges	State description
Instantaneous electrical response	Exact, at $t = 0$	3 (equivalent triangle)	Three effective terminal conductances
All learning trajectories	$0 < \delta \le \delta_0$	$\Theta(\min\{N, L_\delta^2\})$	Finite analytic features of the initial weighted shape distribution
All learning trajectories at saturation	$0 \le \delta \le \delta_N$, fixed N	Exactly $3N$	Distinguishable mass and shape coordinates acquired across preparations and training histories

Lemma 4.1 (Uniform electrical and learning sensitivity)**.** *For every admissible graph H, and all positive initial states g, g' on H with $\sum g_e, \sum g'_e \le 1$,*

$$\|\mathcal{R}_H(g) - \mathcal{R}_H(g')\|_\infty \le K_0 \|\sqrt{g} - \sqrt{g'}\|_2, \qquad K_0 = \sqrt{7}\, e^{105/8}. \tag{4.1}$$

The constant is independent of m, topology, and the smallest conductance.

Proof. Let B_I consist of the interior columns of the incidence matrix, and set $D = \operatorname{diag}(\sqrt{g})$ and $A = B_I^\top D^2 B_I$. If B_I is empty the following projection is zero. Otherwise A is positive definite and

$$\Pi = D B_I A^{-1} B_I^\top D$$

is an orthogonal projector. Its bound $\|\Pi\|_{2\to 2} \le 1$ therefore holds uniformly, including for poorly conditioned A.

For a boundary basis voltage e_a, let u^a be its harmonic extension and $t^a = Bu^a$ the vector of edge voltage drops. The discrete maximum principle gives $|t_e^a| \le 1$. Differentiating the stationary Dirichlet energy with respect to an edge conductance gives, for $a \ne b$,

$$\partial_{g_e} C_{ab}(g) = -t_e^a t_e^b. \tag{4.2}$$

Here and below $C_{ab} = -\Lambda_{ab}$ is one coordinate of Equation (2.1).

The square-root coordinates $x_e = 2\sqrt{g_e}$ flatten the weighted metric introduced in Section 2, with the original time parameter. Uniformity near small positive conductances will follow from the projector bound and the maximum principle. Since $\mathrm{d}g_e = \sqrt{g_e}\,\mathrm{d}x_e$, differentiation of the interior equilibrium equations yields

$$D\,\mathrm{d}t^a = -\Pi \operatorname{diag}(t^a)\,\mathrm{d}x. \tag{4.3}$$

Substituting Equation (4.3) into the derivative of Equation (4.2) gives

$$\nabla_x C_{ab} = -D(t^a \odot t^b), \tag{4.4}$$

$$\nabla_x^2 C_{ab} = -\tfrac{1}{2}\operatorname{diag}(t^a \odot t^b) + \operatorname{diag}(t^b)\Pi \operatorname{diag}(t^a) + \operatorname{diag}(t^a)\Pi \operatorname{diag}(t^b). \tag{4.5}$$

Let $P = \sum_e g_e$ be the total conductance. The projector bound $\|\Pi\| \le 1$ and voltage bound $|t_e^a| \le 1$ imply

$$\|\nabla_x C_{ab}\|_2 \le \sqrt{P}, \qquad \|\nabla_x^2 C_{ab}\|_{2\to 2} \le \tfrac{5}{2}. \tag{4.6}$$

To control these derivatives along the learning trajectory, we also need a bound on the evolving total conductance. Homogeneity of C, Euler's identity, and Equation (2.2) give

$$\dot{P} = C \cdot (q - C) \le \tfrac{1}{4}\|q\|_2^2 \le \tfrac{3}{4}, \quad P(t) \le \tfrac{7}{4} \quad (0 \le t \le 1). \tag{4.7}$$

Equation (4.2) bounds the relative edge speeds, so initially positive conductances remain positive for finite time. In the coordinates $x = 2\sqrt{g}$, the chain rule gives $\dot{x} = -\nabla_x \mathcal{L}_q$. The conductance bound places every trajectory in the convex positive portion of $\|x\|_2^2 \le 7$, where

$$\nabla_x^2 \mathcal{L}_q = (DC)^\top DC + \sum_{a=1}^{3} (C_a - q_a)\nabla_x^2 C_a \succeq -\tfrac{105}{8} I.$$

The inequality follows from $0 \le C_a \le P \le 7/4$, $0 \le q_a \le 1$, and Equation (4.6). It gives a differential comparison for two solutions driven by the same $q(t)$; integrating across the target switches yields $\|x(t) - x'(t)\|_2 \le e^{105t/8}\|x(0) - x'(0)\|_2$. Each output C_a is $\sqrt{7}/2$-Lipschitz on the same positive ball by Equation (4.6). Substitution of $x = 2\sqrt{g}$ proves Equation (4.1), uniformly in the number of target switches. □

Lemma 4.2 (Covering all m-edge replacement networks)**.** *Let $\mathcal{B}_{\le m}$ be the set of future-response maps $\mathcal{R}_H(z)$ produced by all admissible graphs with at most m edges and positive initial conductances of total budget at most one. For $0 < \varepsilon \le 1$,*

$$\mathcal{N}_\varepsilon(\mathcal{B}_{\le m}) \le \Gamma_m \left(1 + \frac{2K_0}{\varepsilon}\right)^m, \qquad \Gamma_m = (m+1)(2m+3)^{2m}. \tag{4.8}$$

Here $\mathcal{N}_\varepsilon$ denotes an internal covering number in the behavior norm Equation (2.4).

Proof. For a fixed graph, $\sqrt{z}$ lies in the positive part of the unit Euclidean ball of $\mathbb{R}^m$. A maximal ε/K_0-separated family in this set has cardinality at most $(1 + 2K_0/\varepsilon)^m$ by the Euclidean ball-packing estimate. It is an internal cover, and Lemma 4.1 maps it to an ε-cover of the graph's responses. Every center corresponds to strictly positive conductances.

An m-edge graph with three named terminals has at most $2m + 3$ relevant vertices. Labeling the vertices and ordering the edges gives an upper bound Γ_m on the number of graph types, including every edge count up to m. The union of the covers for these types proves Equation (4.8). Because this union contains every admissible replacement response, the estimate applies to arbitrary choices of topology and initial conductances. □

5 A quantitatively distinguishable family of sources

To compare the covering estimate with the source family, we construct r^2 independent variations of the initial star shapes and distinguish them by finite experiments with admissible target programs. The proof first identifies observable polynomial moments, then recovers them to sufficient accuracy from finitely many output values. The resulting separation decreases only exponentially with r, which determines the logarithmic scale in the lower bound.

5.1 Observable polynomial moments

Write an individual star as $g = st$, where $s = \sum_e g_e$ is its total conductance and $t = (x, y, 1-x-y)$ describes its shape. This separates a positive scale from a shape in the open positive simplex. The compact set Δ restricts initial shapes; the identities below apply at every positive star state, including shapes outside Δ during learning.

Index the three response coordinates by $a \in \{12, 13, 23\}$, and write $h_a(t) = t_i t_j$ when $a = ij$. The vector fields associated with these coordinates are

$$U_a(g) = \operatorname{diag}(g)\nabla_g S_a(g), \qquad v_{a,\ell}(t) = t_\ell\big(\mathbf{1}_{\{\ell=i\}}t_j + \mathbf{1}_{\{\ell=j\}}t_i - 2h_a(t)\big).$$

Differentiating Equation (2.5) in the scale and shape variables gives

$$U_a s = s h_a(t), \qquad U_a t_\ell = v_{a,\ell}(t). \tag{5.1}$$

On an ensemble, each of these fields acts separately on every star. Their action on weighted polynomial moments is described by the following notation: for a polynomial $p(x, y)$, set

$$O_p(g) = \sum_i s_i p(t_i), \qquad D_a p = h_a p + v_a \cdot \nabla p.$$

The scale and shape identities then give

$$\mathcal{L}_{U_a} O_p = O_{D_a p}, \qquad C_a = O_{h_a}. \tag{5.2}$$

The learning dynamics couple the stars through the common output error. Under a constant target q, their vector field is

$$f_q = \sum_a (q_a - C_a)U_a.$$

Choose four admissible constant targets:

$$q^{(0)} = (1/2, 1/2, 1/2), \qquad q^{(j)} = q^{(0)} + \tfrac{1}{10}e_j \quad (j = 1, 2, 3). \tag{5.3}$$

Their common feedback terms cancel in the difference, so $U_j = 10(f_{q^{(j)}} - f_{q^{(0)}})$ identically as vector fields. Expanding successive Lie derivatives therefore expresses each ordered word in the U_a as a linear combination of ordered words in the admissible closed-loop fields. Every field in this expansion uses the network's own output error.

Lemma 5.1 (Quantitative polynomial observability)**.** *Put $b(x, y) = xy(1 - x - y)$. For every polynomial p of total degree at most k, O_{bp} is a linear combination of ordered Lie derivatives of the three outputs along the four admissible fields in Equation (5.3). If ℓ_w is the derivative order of a word w, the coefficients can be chosen so that*

$$\ell_w \le k + 3, \qquad \sum_w |a_w| \ell_w! \le 48\,000\,000 \cdot 320^k \|p\|_{\mathrm{coeff}}, \tag{5.4}$$

where $\|p\|_{\mathrm{coeff}}$ is the sum of absolute monomial coefficients. The coefficients are independent of N, of the star masses, and of the future target program.

Proof. Appendix A gives the polynomial identities and the full coefficient estimates. The induction starts with b, bx, by, each of which lies in the span of output Lie words. The degree step follows from

$$D_a(bp) = b\big[(t_i + t_j - 5h_a)p + v_a \cdot \nabla p\big]. \tag{5.5}$$

For homogeneous p of degree k, the highest-degree term on the right is $-(2k+5)bh_a^{(2)}p$, where

$$h_{12}^{(2)} = xy, \quad h_{13}^{(2)} = -x^2 - xy, \quad h_{23}^{(2)} = -xy - y^2.$$

Every monomial of degree $k + 2$ can be formed from at most two of these quadratic factors multiplied by a degree-k monomial. After division by $2k + 5$, the remaining terms have degree at most $k + 1$ and coefficient norm at most nine. Starting from a weighted base norm at most 6000, the induction bounds the U-word representation by $6000 \cdot 16^k \|p\|_{\mathrm{coeff}}$, with the derivative factorial $\ell!$ included at each step.

Expanding each U_j as the difference of two admissible fields multiplies the coefficient bound for a word by at most 20^{ℓ_w}. This gives Equation (5.4). The derivative-order convention and factorial estimates are detailed in Appendix A. □

5.2 A cube of independent initial configurations

We next choose a d-dimensional family on which these moments have a quantitative inverse. For an integer $r \ge 2$, let $d = r^2$ and $N \ge d$, and put $k_0 = \lfloor N/d \rfloor$ and $\lambda = k_0/N$. The parameter λ is the total initial mass assigned to each group of repeated shapes, and satisfies

$$\frac{1}{2d} \le \lambda \le \frac{1}{d}. \tag{5.6}$$

Fix the r interior nodes

$$\xi_i = \frac{1}{4} + \frac{i}{12(r+1)} \quad (1 \le i \le r).$$

Assign k_0 stars to each grid position (i, j), with initial shape

$$t_{ij}(u) = (\xi_i + u_{ij}, \xi_j, 1 - \xi_i - \xi_j - u_{ij}), \tag{5.7}$$

Here $u = (u_{ij}) \in [-R_r, R_r]^d$ parameterizes the cube. Assign shape $(1/3, 1/3, 1/3)$ to the other $N - k_0 d$ stars. Every star has initial mass $1/N$. We choose $R_r > 0$ small enough that $t_{ij}(u) \in \Delta$

throughout the cube.

To distinguish the coordinate variations, let l_i be the univariate Lagrange polynomial for $\xi_1, \dots, \xi_r$, and define

$$p_{ij}(x, y) = \frac{(x - \xi_i) l_i(x)^2 l_j(y)}{b(\xi_i, \xi_j)}, \qquad Q_{ij} = b p_{ij}. \tag{5.8}$$

These polynomials satisfy $Q_{ij} = 0$ and $\partial_x Q_{ij}(\xi_a, \xi_b) = \delta_{ia}\delta_{jb}$ at the grid nodes. Thus the initial moment map

$$\mathcal{M}_r(u) = \left(\frac{1}{N} \sum_{\text{all stars}} Q_{ij}(t_\ell(u)) \right)_{i,j} \tag{5.9}$$

has $D\mathcal{M}_r(0) = \lambda I_d$. Controlling the variation of its derivative around this fixed matrix gives a uniform inverse estimate on a small cube. Appendix B gives a choice with $R_r^{-1} \le 2^{36r}$ for which

$$\|\mathcal{M}_r(u) - \mathcal{M}_r(v)\|_\infty \ge \frac{7\lambda}{8} \|u - v\|_\infty, \qquad u, v \in [-R_r, R_r]^d. \tag{5.10}$$

The estimate follows by integrating $D\mathcal{M}_r - \lambda I$ along the segment between the two initial configurations. Comparison with the same matrix throughout the cube controls the difference of the moment values.

5.3 Replacing formal derivatives by finite experiments

The moments in Equation (5.2) are expressed through derivatives at zero episode duration. The response distance, however, compares output values at finite times under admissible programs. We therefore approximate the derivative functionals by linear combinations of such values, controlling the error relative to the distance between source initializations.

Lemma 5.2 (Finite physical discrimination)**.** *For every $r \ge 2$ and every $N \ge r^2$ there is a finite collection $\mathcal{T}_r$ of tests, each consisting of admissible piecewise constant target programs with total duration strictly less than one, common to all initial states in Equation (5.7), such that the vector $O_r(u)$ of observed output values satisfies*

$$\|O_r(u) - O_r(v)\|_\infty \ge \beta_r \|u - v\|_\infty \quad \text{for all } u, v \in [-R_r, R_r]^d, \tag{5.11}$$

where $\beta_r > 0$ is explicit and $\beta_r^{-1} \le 2^{378r}$. The bounds are independent of N.

The construction uses ordered target sequences whose selected episodes have equal positive durations, with other episodes omitted. Varying that common duration supplies one-dimensional interpolation data while preserving the episode order. Polarization then recovers the required mixed coefficients. A uniform analytic estimate for the *difference* of two responses makes the interpolation error proportional to the distance between their starts. Combined with the factorial-weighted bound in Equation (5.4), this gives the exponentially bounded separation in Equation (5.11). The complete proof is in Appendix D, with its constants in Appendix B.

5.4 Comparison with all replacement graphs

Proposition 5.3 (Uniform lower resource bound)**.** *There is a model-dependent $c > 0$ such that for all $N \geq 2$ and sufficiently small $\delta > 0$,*

$$m_{\mathrm{free}}(N, \delta) \geq c \min\{N, [1 + \log(1/\delta)]^2\}. \tag{5.12}$$

Proof. For $r \geq 2$, $d = r^2$ and $N \geq d$, place an $8\delta/\beta_r$-spaced grid in the source cube $[-R_r, R_r]^d$. Lemma 5.2 separates the resulting source responses by at least 8δ. Any replacement responses that approximate them to accuracy δ are therefore separated by at least 6δ. A ball in an internal δ-cover of all m-edge replacement responses can contain at most one of them. Applying Lemma 4.2, with $a_r = R_r\beta_r/4$ and a fixed numerical $B_* > 1 + 2K_0$, gives

$$\left(\frac{a_r}{\delta}\right)^d \leq \Gamma_m \left(\frac{B_*}{\delta}\right)^m. \tag{5.13}$$

If $m \leq d/2$, rearranging this inequality gives $\delta \geq a_r^2/[4B_*(2d + 3)^2]$. The estimates in Appendix B bound this threshold uniformly in r, yielding

$$N \geq r^2, \qquad 0 < \delta \leq 2^{-900r} \quad \Longrightarrow \quad m_{\mathrm{free}}(N, \delta) > \frac{r^2}{2}. \tag{5.14}$$

The covering comparison includes all m-edge replacement networks and hence permits an arbitrary, possibly discontinuous and nonmeasurable, assignment of a replacement to each source initialization.

For $\log_2(1/\delta) \geq 1800$, choose

$$r = \left\lfloor \min\left\{\sqrt{N}, \frac{\log_2(1/\delta)}{900}\right\}\right\rfloor$$

when $N \geq 4$. This choice has $r \geq 2$, so Equation (5.14) gives Equation (5.12) after decreasing the universal constant c to account for integer rounding and the logarithm base. For $N = 2, 3$, the strictly positive initial boundary response excludes a zero-edge replacement at sufficiently small δ. Decreasing c once more covers these cases with the same constant for every N. □

6 A smooth star network achieving the upper bound

We construct a fixed finite ensemble of stars whose positive initial conductances depend smoothly on the source preparation. The proof has three steps. Analytic approximation reduces the responses of one star to finitely many features. A packing construction realizes their averages in a fixed number of stars. A stability estimate then transfers the approximation to the two networks evolving under Equation (2.2), each with its own response error.

6.1 Uniform analytic features of one star

We first prescribe a bounded measurable residual signal $r : [0,1] \to [-1,1]^3$. The resulting normalized star and its response are

$$\begin{aligned} \dot{y} &= V(y,r) := \operatorname{diag}(y) D S(y)^\top r, \qquad y(0) = t \in \Delta, \\ H_{r,t_*,a}(t) &= S_a(y(t_*;t,r)). \end{aligned} \tag{6.1}$$

Here t_* is the observation time, and the argument t of H is the initial shape. Prescribed residuals allow us to compare ensembles under a common forcing. The physical networks will subsequently be compared with their respective residuals $q - C$.

Homogeneity separates a star's initial mass from its shape: a star starting at wt, $w > 0$, follows $wy(\cdot;t,r)$ under the prescribed residual and contributes $wH_{r,t_*,a}(t)$ to the response. Thus an ensemble represented initially by the probability measure $\mu = \sum_j w_j \delta_{t_j}$ has forced response

$$T_\mu[r](t_*) = \int_\Delta H_{r,t_*}(t)\, \mathrm{d}\mu(t). \tag{6.2}$$

Each weight w_j is the sum of the three initial conductances of its star. During training, the star's mass is the sum of its evolving conductances.

Lemma 6.1 (A fixed family of analytic response features)**.** *There exist constants $J \in \mathbb{N}$, $M_{\mathrm{an}}, B_{\mathrm{feat}} > 0$ and C^∞ maps*

$$\varphi_k : \Delta \longrightarrow \mathbb{R}^{D_k}, \qquad D_k = J\frac{(k+1)(k+2)}{2},$$

chosen independently of any forcing r, such that for every $r \in L^\infty([0,1],[-1,1]^3)$, every $0 \le t_* \le 1$, and every response coordinate a, there exists a coefficient vector $b_{r,t_*,a}$ for which

$$\|H_{r,t_*,a} - b_{r,t_*,a} \cdot \varphi_k\|_{L^\infty(\Delta)} \le M_{\mathrm{an}} 2^{-k}, \qquad \|b_{r,t_*,a}\|_1 \le B_{\mathrm{feat}}. \tag{6.3}$$

Proof. For positive y, the star response satisfies $|\partial_{y_i} S_a| \le 1$. Consequently $|\mathrm{d}(\log y_i)/\mathrm{d}t| \le 3$ whenever $\|r\|_\infty \le 1$, and every forced trajectory stays in the common positive box

$$e^{-3}/6 \le y_i(t) \le (2/3)e^3, \qquad 0 \le t \le 1. \tag{6.4}$$

On a sufficiently thin fixed complex neighborhood of this box, the denominator $\sum_i y_i$ remains separated from zero. The rational field $V(y,r)$ and its y-Jacobian are therefore uniformly bounded there for $r \in [-1,1]^3$. Gronwall's inequality keeps a sufficiently small complex perturbation of the two independent initial shape coordinates in this neighborhood up to time one. Picard iteration on short intervals gives holomorphic dependence on those coordinates for every fixed bounded measurable r, with uniform bounds over this forcing class. Compactness of Δ now supplies a common local polydisc radius $\rho > 0$ and a common modulus bound M_0 for all functions $H_{r,t_*,a}$.

Choose finitely many centers $a_1, \dots, a_J$ whose polydiscs of radius $\rho/4$ cover the shape domain, and a fixed smooth partition of unity η_j subordinate to this cover. In the two independent

affine coordinates, define, for $|\beta| \le k$,

$$\varphi_{j,\beta}(t) = \eta_j(t) \left(\frac{t - a_j}{\rho/4} \right)^{\beta}. \tag{6.5}$$

The Taylor coefficients of $H_{r,t_*,a}$ in these rescaled coordinates are bounded by $M_0 4^{-|\beta|}$. Cauchy's estimate bounds the tail by $M_0 2^{-k}$ and the sum of coefficients by $(16/9)JM_0$. Summing with the partition of unity proves Equation (6.3). The same features serve every forcing history and observation time. □

Corollary 6.2 (Finite features control every forced response)**.** *If two probability measures μ, ν on Δ satisfy*

$$\| \int \varphi_k \, \mathrm{d}\mu - \int \varphi_k \, \mathrm{d}\nu \|_\infty \le \varepsilon,$$

then

$$\sup_{\|r\|_\infty \le 1} \sup_{0 \le t \le 1} \|T_\mu[r](t) - T_\nu[r](t)\|_\infty \le 2M_{\mathrm{an}} 2^{-k} + B_{\mathrm{feat}} \varepsilon. \tag{6.6}$$

Proof. Integrate the approximation in Equation (6.3) against $\mu - \nu$, coordinate by coordinate. The two probability measures contribute at most twice the uniform approximation error. The feature mismatch contributes at most ε times the coefficient ℓ^1 norm. □

6.2 A smooth positive packing lemma

A smooth initializer must assign conductances to a fixed list of stars as the desired feature average varies. We begin with continuous barycentric weights on a finite set of candidate shapes. Their support may change, so the stars cannot simply be indexed by the currently positive weights. Instead, cumulative addresses distribute the candidates among a fixed list of slots. Large weights receive separated addresses, while the small weights account for a controlled mixing error. This gives a continuous block map; mollification and a small positive addition then give the required smooth initializer.

The construction takes place in the cone of nonnegative star blocks

$$\mathcal{K}_\Delta = \{0\} \cup \{sx : s > 0,\ x \in \Delta\} = \left\{ z \in \mathbb{R}^3_+ : z_i \ge \tfrac{1}{6} \sum_j z_j \right\}. \tag{6.7}$$

This cone is closed and convex. The final construction will put every block strictly inside the positive conductance region.

Lemma 6.3 (Smooth packing into a fixed number of stars)**.** *Let $\varphi : \Delta \to \mathbb{R}^D$ be smooth on a relative neighborhood, and let $K = \operatorname{conv} \varphi(\Delta)$. For every $\varepsilon > 0$ there exist $M = 2D + 3$ maps*

$$z_1, \dots, z_M : \mathbb{R}^D \longrightarrow (0, \infty)^3,$$

all of class C^∞, such that, for every $v \in K$,

$$\sum_{j=1}^{M} \sum_{e=1}^{3} z_{j,e}(v) = 1, \qquad \frac{z_j(v)}{\sum_e z_{j,e}(v)} \in \Delta, \tag{6.8}$$

$$\Big\| \sum_{j=1}^{M} \Big(\sum_{e} z_{j,e}(v) \Big) \varphi\Big(\frac{z_j(v)}{\sum_e z_{j,e}(v)} \Big) - v \Big\|_\infty \le \varepsilon. \tag{6.9}$$

The number of slots depends only on D; the finite set used to prepare them may grow as the accuracy is tightened.

The construction is proved in Appendix E. Its key estimate charges mixing only to the total mass of small contributions. Large contributions remain separated in the fixed slots. Smoothing the resulting conductance map preserves positivity constraints and total mass; a final arbitrarily small positive addition makes every edge strictly positive.

6.3 Comparison under each network's own feedback

The initial source measure is

$$\mu = \frac{1}{N} \sum_{i=1}^{N} \delta_{t_i}.$$

Apply Lemma 6.3 to φ_k from Lemma 6.1, evaluated at $v = N^{-1} \sum_i \varphi_k(t_i)$. The $M = 2D_k + 3$ resulting blocks initialize M parallel stars. Their $3M$ conductances are strictly positive, sum to one, and depend C^∞-smoothly on the entire source initialization $t \in \Delta^N$. Feature evaluation and packing take place before training. During training, the prepared conductances contain all source-specific state; the network retains no auxiliary feature values or weights.

The source and replacement each use their own output to form the response error. Their trajectories therefore satisfy

$$C_\mu = T_\mu[q - C_\mu], \qquad C_\nu = T_\nu[q - C_\nu]. \tag{6.10}$$

To compare these equations, we need stability with respect to the residual. On the positive box Equation (6.4), the forced response satisfies, for a fixed constant H_{fb},

$$\|T_\nu[r](t) - T_\nu[r'](t)\|_\infty \le H_{\mathrm{fb}} \int_0^t \|r(s) - r'(s)\|_\infty \, \mathrm{d}s. \tag{6.11}$$

The field V has a uniform Lipschitz bound in state and depends linearly on r with uniformly bounded coefficients. Apply Gronwall's inequality to each forced star and then average over the initial probability measure. This proves Equation (6.11) with a constant independent of the number of stars.

For either star ensemble evolving from unit initial mass, Equation (4.7) gives $P \le 7/4$. Each response coordinate satisfies $C_a \le P/4$, so the actual residual belongs to the forcing class: $q - C \in [-7/16, 1]^3 \subset [-1, 1]^3$. Suppose the bound in Equation (6.6) is at most ε_0. Subtract Equation (6.10), compare the two ensembles first at the same residual, and use Equation (6.11) for the difference of residuals. Gronwall's inequality yields

$$\sup_{q \in Q} \sup_{0 \le t \le 1} \|C_\mu(t) - C_\nu(t)\|_\infty \le e^{H_{\mathrm{fb}}} \varepsilon_0. \tag{6.12}$$

Thus the forced approximation controls the trajectories generated by the networks' respective feedback loops.

Proposition 6.4 (Smooth star upper bound)**.** *There are fixed constants $C_1, \delta_1 > 0$ such that, for every $N \ge 2$ and $0 < \delta \le \delta_1$,*

$$m_{\mathrm{star}}(N, \delta) \le C_1 \min\{N, [1 + \log(1/\delta)]^2\}. \tag{6.13}$$

Proof. The initializer from Lemma 6.3 has feature error at most ε. Combining Corollary 6.2 and Equation (6.12) bounds the closed-loop response error by

$$e^{H_{\mathrm{fb}}} \big(2M_{\mathrm{an}} 2^{-k} + B_{\mathrm{feat}} \varepsilon\big).$$

Choose k of order $1 + \log(1/\delta)$ and ε proportional to δ so that each of the two contributions to the displayed bound is at most $\delta/2$. The replacement has

$$3M = 3(2D_k + 3) = 6D_k + 9 = O\big([1 + \log(1/\delta)]^2\big)$$

trainable edges. If $3N$ is smaller, use a copy of the source with the identity initializer. This choice is made from N and δ alone and stays fixed over the whole preparation family and all target histories. Both constructions have smooth initialization and prove Equation (6.13) with a uniform constant. □

Proof of Theorem 3.1 and Corollary 3.2. Choose δ_0 below the thresholds in Propositions 5.3 and 6.4. These estimates and the inclusion of the preparation classes in Equation (2.9) give Equation (3.1). Dividing the upper and lower bounds at the same (N, δ) proves Equation (3.2). □

7 Why exact reproduction requires all source edges

Initially every star has conductance mass $1/N$, leaving only $2N$ freely chosen shape coordinates. Training histories produce different mass redistributions. When preparation and a target prefix are varied jointly, the attained states contain an open $3N$-dimensional set, and subsequent boundary observations distinguish its mass and shape coordinates. A replacement must therefore represent state variations acquired during training as well as those available at preparation. This is a statement about the family of histories: any one fixed target program carries the prepa-

ration manifold to a manifold of the same dimension. The bracket calculation below identifies the additional directions supplied by varying the history.

Lemma 7.1 (Exact observability of a star ensemble)**.** *On the open set where the normalized star shapes t_i are pairwise distinct, the weighted empirical measure*

$$\mu_g = \sum_{i=1}^{N} s_i \, \delta_{t_i}, \qquad g_i = s_i t_i, \tag{7.1}$$

determines its exact future response and is determined by that response. Moreover, near each such state, finitely many admissible target programs and short observation times give a response map of differential rank $3N$ in the local coordinates $(s_i, x_i, y_i)_{i=1}^{N}$.

Proof. The fields U_a and operators D_a in Equation (5.2) apply to arbitrary positive star masses. By the algebraic induction in Lemma 5.1, every moment O_{bp}, with $p(x, y)$ polynomial, is a linear combination of ordered Lie derivatives of the measured response along the four target fields in Equation (5.3).

Suppose two ensembles have equal future outputs under every short admissible target sequence. Differentiation with respect to episode durations shows that their moments O_{bp} agree for every polynomial p. The factor b is positive throughout the open positive shape simplex $x, y, 1 - x - y > 0$. Polynomial interpolation on the finite union of the two shape supports therefore identifies the weighted measures in Equation (7.1). Conversely, equal weighted measures generate equal responses. Stars with the same initial shape follow the same shape trajectory under a common residual, and their masses acquire the same multiplicative factor. This determines the closed-loop response from the measure.

For the differential statement, choose three Hermite polynomials $bp_{i,0}$, $bp_{i,1}$, $bp_{i,2}$ at each distinct shape. Their values and first derivatives select, respectively, its mass and two shape coordinates, with zero value and first derivatives at all other shapes. Squared separation factors followed by a first-derivative correction give these polynomials explicitly, as shown in Appendix C. The Jacobian of the resulting $3N$ moments with respect to (s_i, x_i, y_i) has diagonal blocks $\mathrm{diag}(1, s_i, s_i)$ and hence rank $3N$.

It remains to obtain this rank from finite observations. If their gradients did not span the $3N$ directions, some nonzero tangent vector would annihilate every response observed after a short target sequence. Analyticity in the episode durations would then make it annihilate all mixed derivatives at zero, including the Hermite moments, contradicting their full rank. A finite spanning family of observations therefore has the required rank, with positive episode durations as short as desired. □

Lemma 7.2 (Creation of the missing N directions)**.** *The states reached from interior preparations in Equation (2.6) by admissible target prefixes with positive episode durations and total length less than $1/3$ contain an open subset of the $3N$-dimensional positive state space. The subset may be chosen so that all normalized star shapes are pairwise distinct.*

Proof. At preparation, $s_i = 1/N$ for every star, leaving a $2N$-dimensional manifold of shapes. To find the additional mass directions, evaluate brackets at preparations on the symmetric line

$t = (u, (1-u)/2, (1-u)/2)$, with $u \in (1/4, 1/3)$. The full preparation manifold still allows all shape directions at these points. The combinations $A = U_{12} + U_{13}$ and $B = U_{23}$ preserve the symmetric line, where they take the form

$$A = s\,u(1-u)\,\partial_s + u(1-u)(1-2u)\,\partial_u,$$
$$B = s\,\frac{(1-u)^2}{4}\,\partial_s - \frac{u(1-u)^2}{2}\,\partial_u.$$

Use the bracket convention $[A, B] = AB - BA$ for derivations and write $\mathrm{ad}_A^k B = s a_k(u)\partial_s + b_k(u)\partial_u$. Direct differentiation gives $a_1 = 0$ and

$$b_1 = -\tfrac{1}{2}u^2(u-1)^2, \qquad \begin{aligned} a_{k+1} &= v a_k' - b_k a', \\ b_{k+1} &= v b_k' - b_k v', \end{aligned}$$

where $a = u(1-u)$ and $v = u(1-u)(1-2u)$. Induction gives degree $2k+2$ and leading coefficient $\beta_k = -2^{k-2}(2k-3)!!$ for b_k. For $k \geq 2$, the polynomial a_k has degree $2k+1$ and leading coefficient $(k-1)\beta_k \neq 0$. The distinct degrees make $a_2, \ldots, a_{N+1}$ linearly independent. Consequently there are N distinct points $u_i \in (1/4, 1/3)$ for which the evaluation matrix $[a_{j+1}(u_i)]_{i,j=1}^N$ is invertible.

At this prepared state, the mass components of the N brackets form the matrix $N^{-1}[a_{j+1}(u_i)]$. Together with the $2N$ tangent directions of the preparation manifold, they span the entire state space. By Equation (5.3), the fields A, B belong to the Lie algebra generated by the admissible target fields $f_{q^{(j)}}$. Thus those fields and the preparation manifold jointly supply all $3N$ directions.

We next pass from bracket directions to prefixes with positive durations. Standard commutator conjugations give a finite composition of sufficiently short signed flows whose analytic endpoint map has a nonzero full-rank minor. Regard all episode durations as independent variables in a connected neighborhood of zero. The minor is real analytic and not identically zero, so it cannot vanish throughout the open set of positive durations with total length below $1/3$, however small that set is chosen. Some such positive-duration endpoint map therefore has full rank. The local submersion theorem produces an open set of attained states. Restricting the neighborhood preserves positive conductances and distinct shapes. □

Proof of Theorem 3.3. Apply Lemma 7.1 at a point of the open attained set from Lemma 7.2. After restricting to a smaller nonempty open set U, a fixed finite collection of continuation observations defines a $3N$-component map O that is a local diffeomorphism on U. Choose every continuation to have duration less than $1/3$. Each attaining prefix also lasts less than $1/3$, so their concatenations fit within the common unit horizon.

Suppose replacements with $m < 3N$ edges reproduce every admissible trajectory exactly. For each state in U, choose a source preparation and prefix attaining it, then run the replacement assigned to that preparation through the same prefix. The replacement's current state consists of its graph type and at most m positive conductances. Every continuation response is a smooth function of these conductances. There are finitely many relevant graph types with at most m edges, as in Lemma 4.2. For each type, exhaust the positive conductance domain by countably many compact sets on which the continuation map is Lipschitz. Its image has Hausdorff dimen-

sion at most m, and a finite union of these images cannot contain the open $3N$-dimensional set $O(U)$. This comparison applies to every assignment of initial graphs and conductances, including discontinuous assignments. Hence $m_{\mathrm{free}}(N, 0) \geq 3N$. A copy of the source attains equality, proving Equation (3.3).

For the finite-accuracy assertion, keep N and the continuation observations fixed. After any prefix, Equation (4.7) bounds the replacement's total conductance by $7/4$; throughout a continuation shorter than $1/3$, it is at most two. The proof of Lemma 4.1, applied with these bounds, makes the vector of continuation observations Lipschitz in $\sqrt{z}$, uniformly over graph types. For each fixed graph, this map therefore has a Lipschitz extension to the closed square-root conductance ball with the corresponding budget. This compact closure is used in the dimension argument; the physical replacements still have strictly positive conductances.

Let K_N be the union of the compact extended images for all graph types with at most $3N - 1$ edges. It has Hausdorff dimension less than $3N$. Choose $y \in O(U) \setminus K_N$; compactness gives $\kappa_N = \mathrm{dist}_\infty(y, K_N) > 0$. Take a source preparation and prefix attaining the state with observation vector y. Every smaller replacement, after that same prefix, differs from y by at least κ_N in one of the continuation observations, whatever its initial preparation. Thus $\delta_N = \min\{10^{-3}, \kappa_N/2\}$, together with the source as an upper comparator, proves the asserted equality of the minima. This positive threshold depends on the fixed value of N through the chosen observation map and separation distance. □

8 Discussion

8.1 What is retained by a smaller learning network?

The resource law measures how much conductance state is needed to reproduce a family's future responses. At fixed positive accuracy, an ensemble of representative stars can replace a source of arbitrarily large size. As the tolerance decreases, more distinctions between source organizations become observable and must be retained. For each fixed N, sufficiently stringent accuracy eventually requires all $3N$ edges. The boundary measurements used here therefore distinguish internal organization at a resolution set jointly by the allowed training histories and the response tolerance.

The exact result also separates information present at preparation from state developed during learning. Equal initial star masses leave $2N$ shape coordinates free, but varied training prefixes redistribute the masses in independently observable ways. This is the model-specific content needed to apply the established connection between reachability, observability, and minimal realization [11, 12]. At finite accuracy, the quantitative separation of responses and the graph-independent sensitivity estimate make that connection a resource bound.

Hidden state can affect future observations in other physical settings. Muruga *et al.* demonstrate memory-dependent relaxation and work extraction in a Brownian system coupled to a non-Markovian bath [13]. Here the hidden state is instead the network's evolving conductance organization, and the optimized resource is the number of trainable elements required to reproduce boundary learning trajectories. No work-extraction or energy bound is implied by the edge count.

Related results on resistor networks address different preservation problems. Resource-preserving normalization keeps a designated effective response fixed along admissible rewiring paths and counts internal vertices [14]. Learning-history memory instead concerns changes in subsequent learning after an excursion returns the declared function [15]. Here the graph remains fixed after preparation, the learning law and physical clock are prescribed, and the resource is trainable edge count for reproducing every allowed target history. Neither earlier result supplies the matching finite-accuracy resource bound proved here.

8.2 Physical scope and usable accuracy

The result applies to the specified relative-conductance update and its measurement interface. Coupled-learning experiments [2] show that distributed electrical adaptation is possible, but do not implement the particular gradient law and complete three-terminal response objective studied here. The local voltage products in our law still require shared probes and output residuals. Their generation, communication, and energetic cost must be included in any assessment of an actual device.

The bounds are asymptotic. The explicit sufficient threshold $\delta \le 2^{-900r}$ used in the lower proof is already about 10^{-542} at $r = 2$; it certifies the order of growth, not a practical operating point. The smooth initializer may require a large lookup table and have large derivatives. Preparation cost, measurement precision, delays, energy use, and fabrication errors are outside the optimization. A device asked only to reach the same final response can follow a different path and need a different resource.

Within these assumptions, the matching bounds identify the physical state needed for future-trajectory reproduction to within a constant factor. The two-dimensional analytic shape description achieves this order, and the lower bound shows that changing the passive graph cannot improve its asymptotic scaling. Whether useful constants survive realistic preparation and measurement constraints is a separate physical question.

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Data and code availability

No experimental or numerical simulation dataset is reported. The proofs and finite training constructions supporting the results are included in this article and its appendices. The preprint and its document sources are openly available at doi:10.5281/zenodo.23655709. The archive contains the LaTeX, bibliography, schematic and table files needed to compile the document.

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A Polynomial observation identities and factorial accounting

The same polynomial construction underlies finite-accuracy discrimination and exact observability. We give its base identities, degree induction, and coefficient bounds. In the notation of Section 5, $D_a p = h_a p + v_a \cdot \nabla p$, and the letters $0, 1, 2$ stand for $12, 13, 23$. For a nonempty word $w = (w_1, \dots, w_s)$, define

$$P_w = D_{w_s} \cdots D_{w_1} 1.$$

The identity $D_a 1 = h_a$ makes O_{P_w} an ordered Lie derivative of the measured output C_{w_1}, of order $s - 1$. Thus the constant polynomial in this definition is an algebraic starting point: all measurements begin with a boundary output, and $O_1 = \sum s_i$ is not required as an additional observable.

Expansion of Equation (5.1) over $\mathbb{R}[x, y]$ verifies the base identity

$$\begin{aligned} b = {} & \tfrac{11}{2} P_{01} - \tfrac{5}{2} P_{02} - \tfrac{155}{2} P_{001} + \tfrac{49}{2} P_{002} + \tfrac{157}{2} P_{010} - 4P_{011} + 2P_{012} - \tfrac{19}{2} P_{020} \\ & + 126 P_{0001} - 21 P_{0002} - 147 P_{0010} + 63 P_{0012} - 84 P_{0102}. \end{aligned} \tag{A.1}$$

The degree-one base cases follow from

$$P_{01} = b(1 - 3x), \qquad P_{02} = b(1 - 3y). \tag{A.2}$$

Together, these identities place b, bx, by in the observable span. In Equation (A.1), the sum of absolute coefficients is at most 1000 and every observable derivative has order at most three. Weighting each coefficient by the factorial of its derivative order therefore gives the base bound 6000.

For the induction step, take a homogeneous polynomial p of degree $k \geq 0$. Substitution of $t = (x, y, 1 - x - y)$ gives

$$D_a(bp) = b\big[(t_i + t_j - 5h_a)p + v_a \cdot \nabla p\big]. \tag{A.3}$$

The quadratic parts are $h_{12}^{(2)} = xy$, $h_{13}^{(2)} = -x^2 - xy$, and $h_{23}^{(2)} = -xy - y^2$, while the highest-degree part of v_a is $-2h_a^{(2)}(x, y)$. Euler's identity for p gives the leading term $-(2k + 5) b h_a^{(2)} p$. Every monomial of degree $k + 2$ is obtained from one expression $h_{12}^{(2)} p$, or from a sum of two expressions that isolates $x^2 p$ or $y^2 p$. Dividing by $2k + 5$ leaves a polynomial R of degree at most

$k+1$, whose coefficient norm is bounded by

$$\frac{34+18k}{2k+5} < 9.$$

This bound is uniform in k and supplies the remainder estimate used in Equation (5.5).

Let F_k denote the largest factorial-weighted coefficient sum needed to represent bp, normalized by $\|p\|_{\mathrm{coeff}}$, for polynomials p of degree at most k. The base identities give the safe estimates $F_0, F_1 \le 6000 \cdot 16^1$.

Passing from degree k to $k+2$ appends one derivative operator. The weight increases from $\ell!$ to $(\ell+1)!$, by a factor at most $k+4$, because the previous derivative length is at most $k+3$. The two possible leading terms and division contribute $2/(2k+5)$, and the lower-degree remainder contributes at most $9F_{k+1}$. Hence strong induction gives $F_j \le 6000 \cdot 16^j$, since

$$\frac{2(k+4)}{2k+5} \cdot 16^k + 9 \cdot 16^{k+1} < 16^{k+2}.$$

Finally, substitute $U_a = 10(f_{q^{(a)}} - f_{q^{(0)}})$ in each ordered derivative word. A word of length ℓ increases its coefficient sum by at most 20^ℓ. The bound $\ell \le k+3$ therefore gives

$$6000 \cdot 16^k \cdot 20^{k+3} = 48\,000\,000 \cdot 320^k,$$

which is Equation (5.4). The resulting words involve only the four admissible target fields, with the derivative factorial already included.

B Explicit finite-experiment bounds

We now give the constants used in Lemma 5.2 and Equation (5.14). The estimates are chosen for simplicity and uniformity in N; they give a sufficient theoretical accuracy threshold.

B.1 Grid norms and inverse Jacobian

Set $d = r^2$ and use the equally spaced nodes $\xi_i = 1/4 + i/[12(r+1)]$. The denominator in the Lagrange polynomial $l_i(x)$ contains the factor $(i-1)!(r-i)!$. Combining the binomial estimate with $(r-1)! \ge ((r-1)/e)^{r-1}$ yields, for $r \ge 2$,

$$\|l_i\|_{\mathrm{coeff}} \le \frac{[16(r+1)]^{r-1}}{(i-1)!(r-i)!} \le 9 \cdot 96^{r-1} \le H_r := 256^r. \tag{B.1}$$

The polynomials p_{ij} in Equation (5.8) have degree at most $k = 3r-2$. Define

$$P_r = 64H_r^3, \quad J_r = 192(3r+1)^2 H_r^3, \quad R_r = \min\left\{\frac{1}{48(r+1)}, \frac{1}{8dJ_r}\right\}, \tag{B.2}$$

Then $\|p_{ij}\|_{\mathrm{coeff}} \le P_r$ and $\|\partial_{xx} Q_{ij}\|_{\mathrm{coeff}} \le J_r$. Each coordinate u_{ij} changes only its corresponding group of repeated shapes. Summing the derivative variations along one row therefore gives

the bound $\lambda d J_r R_r \le \lambda/8$ for the maximum row-sum norm of $D\mathcal{M}_r - \lambda I$ on the cube. Integrating along a segment, with the fixed matrix λI as reference, proves Equation (5.10). The other constraint on R_r keeps all initial shapes inside the prepared simplex.

B.2 The uniform complex tube

In the coordinates $y_i = Ng_i$, the source field is given by Equation (D.1). Its only possible singularities occur when a star's three conductances sum to zero. On the complex ℓ^∞ tube of radius $1/24$ around Δ^N,

$$|y_{ie}| \le \tfrac{17}{24}, \qquad |\sum_e y_{ie}| \ge \tfrac{7}{8},$$

so the field is analytic there. Direct differentiation of S gives

$$|S_a(y_i)| < 1, \quad |\partial_e S_a(y_i)| < 2, \quad |\partial_{ef} S_a(y_i)| < 16.$$

For each admissible constant target, these estimates imply $\|f\|_\infty < 9$ and $\|Df\|_{\infty\to\infty} < 512$ for the normalized field. Differentiating the feedback produces N summands with a factor $1/N$ each, so both estimates are uniform in N.

An ordered flow composition whose absolute complex episode durations sum to at most $\tau = 1/1024$ remains inside the tube: its displacement is bounded by $9\tau < 1/48$, leaving a strict first-exit margin. The derivative with respect to the initial state is bounded by $e^{512\tau} < 2$, and the output derivative has row sum at most six. Integrating these derivatives along the segment between two admissible starts gives Equation (D.2) with coefficient 16. This argument controls the output difference proportionally to the initial-state difference.

B.3 Univariate interpolation and the factorial balance

Let F be holomorphic and bounded by B on the shifted Bernstein ellipse of parameter 64 around $[0, 1]$. Its Chebyshev coefficients satisfy $|c_j| \le 2B \cdot 64^{-j}$. The three-term recurrence bounds the coefficient ℓ^1 norm of $T_j(2s-1)$ by 8^j. At the Chebyshev–Lobatto nodes, each mode T_j aliases to a mode of degree at most j, while degrees $j \le M$ are reproduced exactly. Therefore every coefficient $[s^\ell]$, $0 \le \ell \le M$, satisfies

$$|[s^\ell](F - I_M F)| \le 4B \sum_{j>M} (8/64)^j = \frac{4}{7} B \cdot 8^{-M}.$$

The Chebyshev coefficients of $I_M F$ are at most twice the largest nodal value. Hence its monomial coefficients are bounded by $2\sum_{j=0}^{M} 8^j \max_{\text{nodes}} |F| \le 32^M \max |F|$ for $M \ge 1$, proving Equation (D.4). Both estimates concern coefficient extraction directly.

Take $a = 1/65536$ and set $t = as$. For s on the shifted ellipse of parameter 64, $|t| \le a(2+64+1/64)/4 < \tau$. The polarization identity Equation (D.3) has at most 2^ℓ terms, and $\ell^\ell \le 3^\ell \ell!$. Including the scale factor $a^{-\ell}$ gives the bounds used after Equation (D.4): $\ell!(6/a)^\ell \cdot 32^M$ for the norm of the nodal-value functional and $16\ell!(6/a)^\ell \cdot 8^{-M}\|u - v\|_\infty$ for its error on the difference of two starts. Each test executes the remaining positive-duration episodes in their

fixed chronological order, omitting the zero-duration episodes.

B.4 A uniform explicit threshold

With the polynomial degree $k = 3r - 2$, collect the preceding bounds as follows:

$$W_r = 48\,000\,000 \cdot 320^k P_r (6/a)^{k+3}, \qquad M_r = 50r, \tag{B.3}$$

$$A_r = W_r \cdot 32^{M_r}, \qquad \beta_r = (8dA_r)^{-1}, \tag{B.4}$$

$$a_r = R_r \beta_r / 4, \qquad B_* = 3^{16}. \tag{B.5}$$

Combining polynomial observability with the interpolation estimate represents each grid moment by a linear combination of finite test observations with coefficient ℓ^1 norm at most A_r. When applied to a difference of source starts, the error of this representation is bounded by

$$16 W_r \cdot 8^{-M_r} \|u - v\|_\infty.$$

For every $r \geq 2$, the choice of interpolation degree gives $8^{M_r} \geq 256 d W_r$, so the error is at most $\|u - v\|_\infty/(16d)$. The separation coefficient in Equation (5.10) satisfies $7\lambda/8 \geq 7/(16d)$. Subtracting the error and dividing by A_r therefore proves Equation (5.11) with the stated β_r.

To evaluate the entropy comparison, suppose $m \leq d/2$. Then $d/(d-m) \leq 2$, $m/(d-m) \leq 1$, and

$$\Gamma_m^{1/(d-m)} \leq (d+1)^{2/d} (2d+3)^2 \leq 4(2d+3)^2.$$

These inequalities, together with $a_r < 1$, allow Equation (5.13) to be rearranged as

$$\delta \geq \frac{a_r^2}{4 B_* (2d+3)^2}.$$

Consequently, the smaller threshold

$$\eta_r = \min\left\{10^{-3}, \frac{a_r^2}{8 B_* (2d+3)^2}\right\} \tag{B.6}$$

forces $m > d/2$ whenever $\delta \leq \eta_r$.

It remains to express this threshold as an exponential in r. The definitions in Equations (B.1) to (B.3) give, for $r \geq 2$,

$$W_r \leq 2^{125r}, \quad A_r \leq 2^{375r}, \quad R_r^{-1} \leq 2^{36r}, \quad \beta_r^{-1} \leq 2^{378r}, \quad a_r^{-1} \leq 2^{415r}, \quad \eta_r^{-1} \leq 2^{852r}. \tag{B.7}$$

Indeed, $48\,000\,000 < 2^{26}$, $320 < 2^9$, $H_r = 2^{8r}$, and $6/a < 2^{19}$ imply $\log_2 W_r \leq 108r + 33 \leq 125r$. The choice $M_r = 50r$ adds $250r$ to $\log_2 A_r$ and gives $8^{M_r} = 2^{150r} \geq 256 r^2 \cdot 2^{125r}$. Also,

$$8dJ_r \leq 2^{15 + 4\log_2 r + 24r} \leq 2^{36r}, \qquad 8B_*(2r^2+3)^2 \leq 2^{33 + 4\log_2 r} \leq 2^{22r}.$$

Substitution gives the remaining bounds in Equation (B.7), including $10^3 = (10^{-3})^{-1}$, the contribution of the first term in the minimum defining η_r to the bound on η_r^{-1}, for $r \geq 2$. In particular,

$\eta_r \geq 2^{-900r}$, which proves the sufficient threshold in Equation (5.14).

C Explicit Hermite observability and the orbit argument

The constructions below establish the observation and reachability ranks used in Lemmas 7.1 and 7.2 for every finite N.

Let $t_1, \dots, t_N$ be distinct points of the open positive shape simplex, expressed in two independent affine coordinates. Define

$$L_i(t) = \prod_{j \neq i} \frac{\|t - t_j\|_2^2}{\|t_i - t_j\|_2^2}$$

using the Euclidean norm in these coordinates. The factor L_i equals one at t_i and vanishes together with its gradient at every t_j, $j \neq i$. For $a = 1, 2$, set

$$p_{i,a}(t) = \frac{(t_a - t_{i,a})L_i(t)}{b(t_i)},$$
$$p_{i,0}(t) = \frac{L_i(t)}{b(t_i)} \left[1 - \left(\nabla \log b(t_i) + \nabla L_i(t_i)\right) \cdot (t - t_i)\right].$$

Then $bp_{i,0}(t_j) = \delta_{ij}$ and all its first derivatives at the shape points vanish. The polynomial $bp_{i,a}$ vanishes at every shape point; its gradient is e_a at t_i and zero at the others. Differentiating the moments $\sum_j s_j bp(t_j)$ gives a block-diagonal Jacobian with blocks $\mathrm{diag}(1, s_i, s_i)$, each invertible for positive masses. By Appendix A, finite-length Lie words express all these moments. Analytic dependence on the durations of admissible target episodes then transfers the full differential rank to a finite collection of response observations.

For reachability, use $a(u) = u(1-u)$ and $v(u) = u(1-u)(1-2u)$. The recurrence in Lemma 7.2 gives the leading coefficients $\beta_1 = -1/2$ and

$$\beta_{k+1} = 2(2k-1)\beta_k, \qquad \beta_k = -2^{k-2}(2k-3)!!.$$

Because $\deg b_k = 2k+2$, the recurrence for a_{k+1} has leading term $k\beta_{k+1}u^{2k+3}$ for $k \geq 1$, starting from $a_1 = 0$ and $a_2 \neq 0$. The polynomials $a_2, \dots, a_{N+1}$ consequently have strictly increasing odd degrees and nonzero leading coefficients. In every nonempty open subinterval of $(1/4, 1/3)$, one can choose N distinct evaluation points giving an invertible matrix: failure for every such choice would force a nonzero linear combination of the polynomials to vanish on the whole interval. These evaluations provide all N mass directions.

The positive-duration conclusion follows from the local orbit and submersion argument. Conjugated finite flows with signed times recover the bracket tangent directions and yield an analytic endpoint map with a nonzero full-rank minor. With episode durations treated as independent parameters, the positive durations of sufficiently small total length form an open subset of the same connected parameter neighborhood. A nonzero analytic minor cannot vanish on all of that subset. Some positive-duration endpoint map therefore has full rank. The resulting open set is attained by varying both the source preparation and the initial target sequence.

D From ordered derivatives to finite observations

The following proof supplies the finite experiments used in Lemma 5.2. The numerical estimates it invokes are derived in Appendix B.

Proof of Lemma 5.2. Normalize the conductances by $y_i = Ng_i$. The coupled source field becomes

$$\dot{y}_i = \operatorname{diag}(y_i) D S(y_i)^\top \left[q - \frac{1}{N} \sum_j S(y_j) \right]. \tag{D.1}$$

The shared feedback is the average $N^{-1} \sum_j S(y_j)$, which permits bounds independent of the number of stars. On the complex tube of radius $1/24$ around the admissible real starts, the field and its Jacobian satisfy $\|f\|_\infty < 9$ and $\|Df\|_{\infty\to\infty} < 512$. Consequently, ordered compositions of the four admissible fields are holomorphic in their episode durations whenever the sum of absolute complex durations is at most $\tau = 1/1024$. Their derivatives with respect to the initial coordinate u are uniformly bounded. Applied to two observed outputs $G_u(z) - G_v(z)$, these bounds give

$$|G_u(z) - G_v(z)| \le 16\|u - v\|_\infty, \qquad \sum_j |z_j| \le \tau. \tag{D.2}$$

The factor $\|u - v\|_\infty$ is needed to retain a uniform separation estimate even for arbitrarily close initial states. The complex-tube and variation bounds are derived in Appendix B.

For an ordered word of length ℓ, let $G(z_1, \dots, z_\ell)$ be the output of the corresponding flow composition, with the episode order fixed. For each subset $S \subset \{1, \dots, \ell\}$ write $u_S = \mathbf{1}_S/\ell$. Coefficient polarization gives

$$\partial_1 \cdots \partial_\ell G(0) = \ell^\ell \sum_{S \subset \{1,\dots,\ell\}} (-1)^{\ell - |S|} [t^\ell] G(t u_S). \tag{D.3}$$

Inclusion–exclusion removes every degree-ℓ monomial that omits a variable, leaving $z_1 \cdots z_\ell$. Each exponent is one, so its coefficient is the mixed derivative on the left. In each test, the nonzero-duration episodes occur in their original chronological order; a zero coordinate $u_{S,j}$ simply omits episode j.

Take $a = 1/65536$. Scaling by a maps the Bernstein ellipse of parameter 64 inside $\sum |z_j| < \tau$, so each function $s \mapsto G(a s u_S)$ is analytic there. Let I_M be its degree-M Chebyshev–Lobatto interpolant on $s \in [0, 1]$. If the function is bounded by B on this ellipse, then, for $\ell \le M$,

$$|[s^\ell](G - I_M G)| \le \tfrac{4}{7} B \cdot 8^{-M}, \qquad |[s^\ell] I_M G| \le 32^M \max_{\text{nodes}} |G|. \tag{D.4}$$

The first estimate bounds the coefficient error by summing aliased Chebyshev coefficients. The second bounds the corresponding linear functional of the observed nodal values. Both are derived in Appendix B. Here $[s^\ell]$ extracts a monomial coefficient; conversion to the ℓth derivative would introduce a factor $\ell!$, which is unnecessary for Equation (D.3).

Apply Equation (D.4) to the difference of two source responses, with $B = 16\|u - v\|_\infty$ from Equation (D.2). The polarization formula and $\ell^\ell \le 3^\ell \ell!$ give a finite observation functional of norm at most $\ell!(6/a)^\ell \cdot 32^M$, whose error is at most $16\ell!(6/a)^\ell \cdot 8^{-M}\|u - v\|_\infty$. The factorial

weight in Equation (5.4) controls the sum of these bounds over all words representing a grid moment.

With $M = 50r$, the combined error is smaller than a fixed fraction of the separation in (5.10). Appendix B gives the constants. Dividing by the norm of the observation functional yields (5.11). This construction uses finitely many tests, each with nonnegative episode durations summing to at most $a < 1$. □

E Smooth positive realization in fixed slots

We prove Lemma 6.3 by assigning a continuous block map to fixed slots before smoothing it.

Proof of Lemma 6.3. Write $Q = D + 1$. Select finitely many candidate shapes $x_a \in \Delta$ whose feature images form an $\varepsilon/4$-net of $\varphi(\Delta)$ in Euclidean norm. Their convex hull $P = \operatorname{conv}\{\varphi(x_a)\} \subseteq K$ is within $\varepsilon/4$ of every point of K. The Euclidean projection $\pi_P : \mathbb{R}^D \to P$ is globally Lipschitz. Triangulate P using its existing vertices, working in its affine hull if necessary. The resulting continuous piecewise affine nodal weights $w_a(y)$ satisfy

$$w_a(y) \geq 0, \quad \sum_a w_a(y) = 1, \quad \sum_a w_a(y)\varphi(x_a) = y,$$

with at most Q positive weights at each y. Composition with π_P gives globally Lipschitz weights $w_a(v)$ with the same sparsity bound and feature average $\pi_P(v)$.

We next assign addresses to these weights in a fixed index order. Set

$$\theta = \min\left\{\frac{1}{2}, \frac{\varepsilon}{8Q \max\{1, \operatorname{Lip}\varphi\}}\right\} > 0.$$

Choose the C^1 cutoff $\sigma(t)$ equal to zero for $t \leq 0$, to $3t^2 - 2t^3$ for $0 < t < 1$, and to one for $t \geq 1$. The cutoff records a full unit for each weight at least θ and varies continuously as a smaller weight appears or disappears. Define the cumulative addresses by

$$\chi_a = \sigma(w_a/\theta), \qquad c_a = 2\sum_{b<a} \chi_b + \chi_a. \tag{E.1}$$

At most Q cutoffs are positive, so $0 \leq c_a \leq 2Q$. These addresses are globally Lipschitz functions of v. To distribute each candidate between adjacent slots, let $\beta_j(c)$, $j = 0, \ldots, 2Q$, be the C^1 partition of unity on $[0, 2Q]$ with $\beta_j = 1 - \sigma(c - j)$ and $\beta_{j+1} = \sigma(c - j)$ on $[j, j + 1]$, and all other terms zero there. The $M = 2Q + 1 = 2D + 3$ preliminary blocks are

$$z_j^0(v) = \sum_a w_a(v)\beta_j(c_a(v))x_a, \qquad j = 0, \ldots, 2Q. \tag{E.2}$$

The blocks are continuous in v, belong to $\mathcal{K}_\Delta$, and have total mass one. Two large weights, $w_a, w_b \geq \theta$ with $a < b$, have addresses separated by

$$c_b - c_a = \chi_a + \chi_b + 2\sum_{a<\ell<b} \chi_\ell \geq 2.$$

Since $\beta_j(c) > 0$ only for $c \in (j-1, j+1)$, each slot receives a positive contribution from at most one large weight. The total mass of all small weights is at most $Q\theta$. A vanishing weight contributes a vanishing block in Equation (E.2), while the labels of the physical slots remain fixed.

Let $\nu = \sum_a w_a \delta_{x_a}$, and let ν^0 be the measure formed from the masses and normalized shapes of the nonzero blocks z_j^0. Couple each fragment $w_a \beta_j(c_a)$ to the barycenter of its destination slot. This gives

$$W_1(\nu, \nu^0) \le 2Q\theta \operatorname{diam}_{\ell^1}(\Delta) = 2Q\theta. \tag{E.3}$$

To obtain the bound, consider a slot's possible large contribution. Its displacement to the barycenter costs at most the small mass in that slot times the diameter of Δ. Moving the small contributions costs at most the same amount again. Convexity of Δ keeps every barycenter in the allowed shape set. The feature error is therefore at most $2Q\theta \operatorname{Lip}\varphi \le \varepsilon/4$, in addition to the projection error of at most $\varepsilon/4$.

We now smooth the block map and make every coordinate positive. The map $v \mapsto (z_j^0(v))_j$ has a finite global Lipschitz constant Λ when its range is measured by the sum of block ℓ^1 norms. To estimate the effect of perturbing blocks, extend the feature map homogeneously to $\mathcal{K}_\Delta$:

$$\widehat{\varphi}(z) = \begin{cases} (\sum_e z_e)\varphi(z/\sum_e z_e), & z \neq 0, \\ 0, & z = 0. \end{cases}$$

Writing $M_\varphi = \sup_\Delta \|\varphi\|_\infty$ and separating the changes in mass and shape gives

$$\|\widehat{\varphi}(z) - \widehat{\varphi}(z')\|_\infty \le (M_\varphi + 2\operatorname{Lip}\varphi)\|z - z'\|_1. \tag{E.4}$$

Summing this inequality controls the total feature change by the sum of block perturbations with the same constant.

Convolve the global block map with a nonnegative C^∞ mollifier of radius $\alpha > 0$ chosen so that $\Lambda\alpha \le \varepsilon/[4(1 + M_\varphi + 2\operatorname{Lip}\varphi)]$. Convolution preserves the convex constraints $z_j \in \mathcal{K}_\Delta$ and $\sum_{j,e} z_{j,e} = 1$. The resulting blocks z_j^α are smooth, and (E.4) bounds their additional moment error by $\varepsilon/4$. For strict positivity, set $x^\circ = (1/3, 1/3, 1/3)$ and choose

$$\vartheta = \min\left\{\frac{1}{2}, \frac{\varepsilon}{8(1 + M_\varphi + 2\operatorname{Lip}\varphi)}\right\}, \qquad z_j(v) = (1-\vartheta) z_j^\alpha(v) + \frac{\vartheta}{M} x^\circ.$$

Every coordinate now satisfies $z_{j,e} \ge \vartheta/(3M) > 0$. The total mass is still one, and every normalized shape lies in Δ. This final perturbation costs at most $\varepsilon/4$ in feature error. Adding the projection, packing, smoothing, and positivity errors proves Equation (6.9). The resulting map is smooth on all of $\mathbb{R}^D$ and uses exactly M slots. □

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